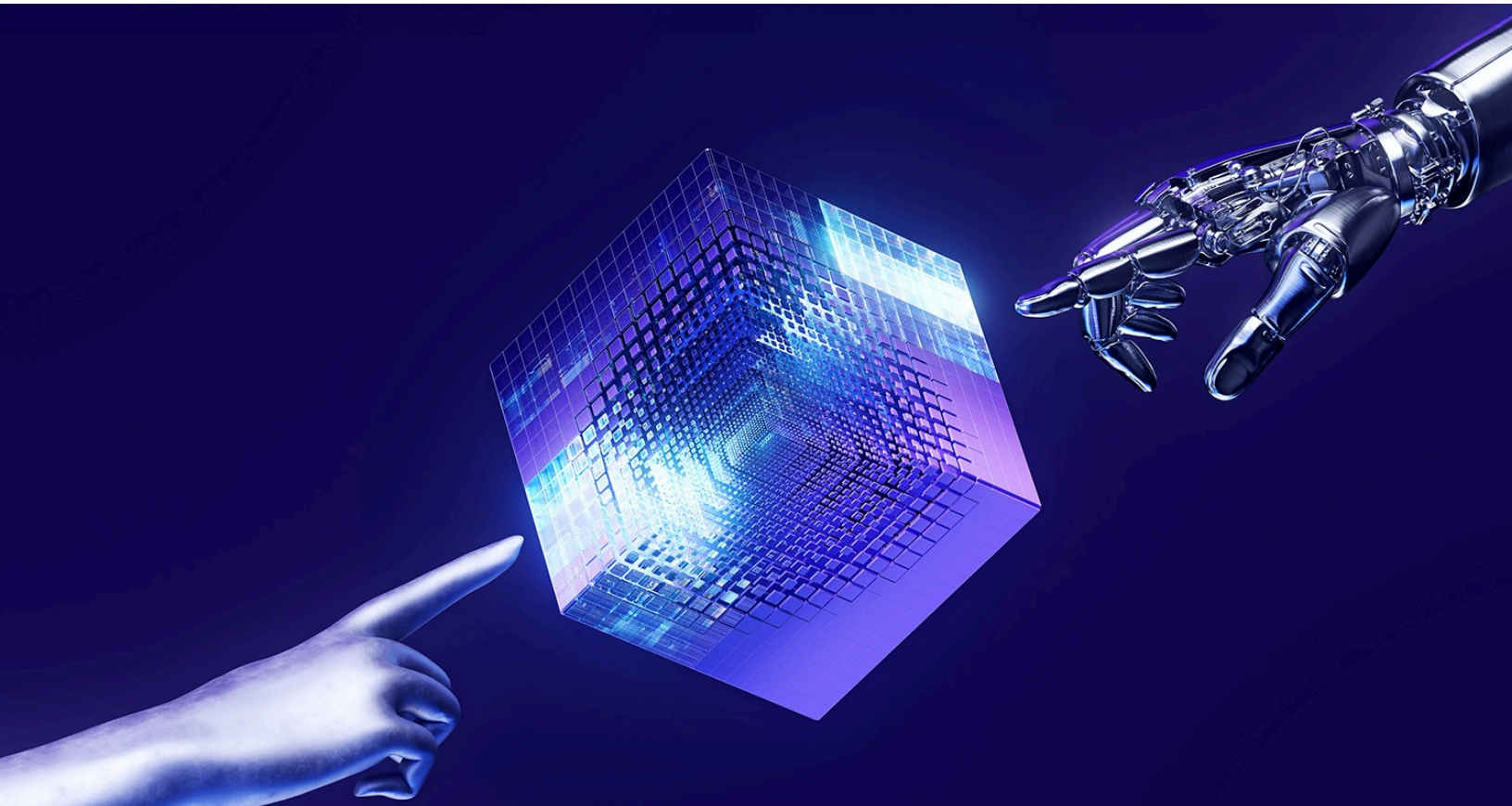


McKinsey Technology

Designing an end-to-end technology workforce for the AI-first era

CIOs are rebuilding their teams for the agentic AI age. This requires rethinking how to hire, build internal capabilities, and negotiate vendor strategies to speed innovation.

*by Henning Soller and Mike Stefanelli
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AI is forcing companies to redesign their technology organizations from the inside out. As agents take on more work, companies must rebalance how they approach tech hiring, internal capability building, and vendor negotiation. The end goal is to generate real ROI from AI, but getting the formula right is anything but simple.

For chief information officers (CIOs), it requires becoming architects of change in close collaboration with business leaders. Our research finds that two-thirds of top-performing companies have technology leaders “very involved” in [crafting enterprise strategy](#), compared with 52 percent of other organizations.¹ Together, CIOs and CEOs—working closely with chief human resource officers—can reshape their organizations to empower teams and generate maximum value.

Leaders can start by asking themselves three important questions: What new technology talent should we hire? Which capabilities should we help our employees build versus training agents to deploy them? Which type of vendor partnerships can deliver real ROI? This article will help CIOs chart a course forward, providing them with actionable tactics they can use to ensure their talent, skill development, and partnership strategies bring their companies into the agentic age.

Pressure from multiple fronts

Companies everywhere and in every sector must fundamentally rethink their technology organizations—and do so while facing four mounting pressures. Tech leaders are expected to cut overall costs while delivering more innovation. They must scale global delivery while [managing increasing geopolitical risk](#). They are tasked with ensuring that partnerships with vendors deliver ROI, not just empty promises. And they must hire new talent while reskilling existing workforces—helping every employee become a technologist—while empowering human–agent teams to deliver AI at scale.

Let’s look at each of these four pressures in more detail to understand why CIOs are walking a tightrope today.

Cost pressures

In the past two years, the business environment has tightened. [CIOs are facing increased pressure to reduce spending](#) on “run” costs—the spending needed to keep tech infrastructure operational—to free up budget for innovative “change” initiatives such as AI rollouts. But run costs are rising, too, since AI requires robust infrastructure. That’s why almost every company is spending more on enterprise technology—and especially top-performing companies.² A quarter of these companies plan to increase their technology budgets by more than 10 percent this year, compared with just 3 percent of other companies.³

Geopolitical risks

Geopolitical uncertainty further complicates the picture. [Global capability centers](#) have evolved from cost-focused delivery hubs into engines of innovation and product development. Yet this

¹ “[McKinsey Global Tech Agenda](#),” McKinsey, February 9, 2026.

² Top performers are defined as those creating significant value; these organizations have seen an average growth rate of at least 10 percent in both revenue and EBIT over the past three years.

³ “[McKinsey Global Tech Agenda](#),” McKinsey, February 9, 2026.

evolution brings new risks, including regulatory exposure, data security concerns, and decisions on where to hire. In this context, offshoring decisions that once seemed straightforward now require a more nuanced view of enterprise resilience and total cost of ownership.

Shifting vendor value chains

AI has forced a moment of reckoning between companies and their technology vendors. The software industry is entering a new shift where buyers and incumbent vendors alike must reimagine their value propositions, technology stacks, and operating models for an AI-centric era. For buyers, this includes integrating vendors and off-site teams more deeply into the company's AI-native platforms, as well as [scaling internal development to embed intelligence across workflows](#) and products.

New workforce dynamics

The war for AI talent has never been fiercer, compelling companies to compete heavily for the most in-demand technical hires. On the flip side, agentic AI is doing more lower-level development work, engendering layoffs and requiring reskilling of remaining tech employees. [High-impact employees are those who master gen AI as a superpower](#) to deliver true transformation for their companies. As these competing factors play out, CIOs aren't sure how to hire for today, let alone for the next three to five years.

What CIOs must do despite these pressures

The pressures facing technology leaders are real, but they do not remove the need for decisive action. If anything, they narrow the margin for error. CIOs who succeed in this environment do not attempt to solve every constraint at once. Instead, they focus on a small set of choices that shape how value is created as AI takes on more execution work. Those choices center on three areas: *how technology talent is hired, how internal capabilities are built and sustained, and how vendor relationships are structured and governed*. Getting these moves right is not easy, especially since CIOs will continue to face cost pressures and resiliency risks. But putting an effective three-part plan in place can determine whether AI investments deliver lasting value rather than short-lived efficiency gains. And with the right balance, companies can ultimately [triple the return they get from their technology investments](#).

Rethinking technology hiring in an agentic world

As agentic AI takes on more routine development and maintenance work, the logic of technology hiring is evolving. The central question is no longer how to scale engineering capacity but how to allocate human talent where judgment and decision-making still matter. This requires hiring some AI specialists but also reskilling existing teams.

Leading organizations are hiring fewer technologists overall, but they are being far more selective about who they bring in. Demand is rising for senior engineers, architects, product managers, and designers who can define standards and orchestrate development work across

internal and external teams, vendors, and agents. At the same time, the business case for hiring large numbers of junior developers is weakening as agentic AI absorbs much of the work that once justified those roles.

The productivity implications of the future [agentic organization](#) are significant. In our experience, with agents as tools, expert technologists can be several times more productive than less experienced peers, and the pay differential between them is modest relative to the output gap. Companies that continue to hire for volume rather than expertise risk inflating costs without increasing impact.

Several organizations have already begun to adjust. A global healthcare company reorganized hundreds of technologists into a [product-based operating model](#), accelerating delivery and creating clearer ownership of technology outcomes. As part of this shift, hiring priorities moved away from generic engineering roles toward profiles with deep platform and architectural expertise. This approach also supported early gen AI pilots by ensuring that human talent was focused on high-value design and oversight rather than routine execution.

The best CIOs aren't hiring for today, but for tomorrow. Hiring strategies must reflect not just today's skill gaps but how the nature of technology work will evolve in the next three to five years. In an agentic world, fewer hires can deliver more value if they are placed deliberately.

Building internal capabilities that AI cannot replace

Hiring alone is not enough. As agents take on more execution work, companies must also decide which capabilities they will deliberately build and sustain internally. These decisions have always defined the long-term shape of the technology organization—but never more so than today. Successful companies will be those that infuse agentic AI throughout their organizations to create [a true hybrid human–agent workforce](#). Equipping teams with the technology skills they need to thrive in an agentic world and deploying effective change management to ensure that people embrace this shift are critical for long-term competitiveness. People will need support to step into new types of [roles where they manage agents alongside colleagues](#).

The most effective CIOs are taking a skills-based view of their workforce rather than relying on static role definitions. They identify the capabilities that differentiate each team, assess current proficiency levels, and invest systematically in development. This often reveals that the constraint is not head count but clarity on where untapped skills lie. Without a shared understanding of which skills matter most—and top-down [change management to communicate how agents can be coworkers](#)—reskilling efforts tend to remain ineffective.

The global healthcare company illustrates what a disciplined approach to skills development could look like. After transitioning to a [product and platform operating model](#), the company made capability building a strategic priority. Over several months, it identified many hundreds of roles with critical skills across product management, full-stack engineering, and design. These

people could form the core team to deliver on the company's technology strategy. For the majority of those roles, the company mapped skill proficiency, finding that half of people were proficient in their core skills and analyzing who used these skills most day-to-day. This clarity enabled the company to do targeted development, define career paths, and eliminate overlapping roles. It also engaged in more skill-based hiring to fill critical needs. The changes could result in tens of millions of dollars in new value creation over the next few years.

Companies cannot outsource their understanding of how AI impacts their development road maps, platforms, data, or architecture without losing control of value creation. At the same time, they should not attempt to build every capability in-house. The challenge for CIOs is to draw that line explicitly and revisit it as AI capabilities mature.

Renegotiating the role of vendors

As AI reshapes internal work, it is also creating an upheaval in the economics of vendor relationships, both outsourced development contracts and software-as-a-service (SaaS) portfolios. Leading organizations are rethinking how they engage vendors, shifting from managed services to managed value.

Outsourced development contracts increasingly link compensation to results such as delivery speed, quality, modernization progress, or business impact. Older contracts were designed for a labor-intensive delivery model. They rewarded activity, head count, or volume rather than outcomes. In an agentic environment, where development becomes far faster and nimbler under the guidance of a few experienced engineers, this misalignment becomes costly. Companies increasingly want their vendor to deploy automation and AI tools rather than be constrained by rigid scope definitions. They want governance to focus on joint performance and continuous improvement rather than oversight alone.

Vendors are moving from seat-based SaaS consumption models toward API-first, outcome-linked software and agentic workflows. For buyers, this shift is driving vendor consolidation, deeper gen AI alliances designed for interoperability, and multiprovider architectures. As vendors move to outcome-linked models, buyers are looking to reduce vendor lock-in and accelerate internal capability building, while deepening partnerships with their key partners.

A bank that moved to a product platform operating model provides one example of successful vendor renegotiation. Traditionally, the bank's operational model enforced central decisions that were not aligned with technology product strategy. Product teams took direct ownership of vendor selection and strategic relationship building, aligning vendor decisions to individual product road maps to accelerate overall delivery. A central KPI was introduced to ensure overall outcomes across products. Additionally, targets for AI-driven automation were negotiated for every strategic vendor. The result was a 10 percent productivity uplift across both internal and external teams.

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For CIOs, vendor negotiation is no longer an annual procurement exercise. It's a continuous strategic practice that can bring technology costs under control, while also creating entirely new enterprise value. As AI absorbs more execution work, the value of vendors lies less in capacity or solution delivery and more in their ability to cocreate measurable outcomes.

The actions outlined above will also carry several additional implications that CIOs will need to keep in mind as they move forward. Perhaps most significantly, the shift is not purely technological. It also represents a profound [change management challenge](#). Organizations will need to help employees navigate fast-evolving roles, while leaders will be asked to guide teams through transformation even as the end state continues to emerge. In many cases, this will require reimagining roles rather than simply hiring or upskilling for existing ones. At the same time, CIOs will need to begin thinking through [how hybrid teams of humans and AI agents will operate](#) in practice, including the new collaboration models and governance structures needed to make these systems effective and sustainable.

Agentic AI is reshaping how companies operate. That means CIOs can no longer treat hiring, capability building, and vendor strategy as separate decisions. These levers now reinforce one another. Hiring determines where human judgment sits in the organization. Capability building determines whether AI amplifies that judgment or bypasses it. Vendor relationships determine whether productivity gains accrue to the enterprise or are absorbed by outdated delivery models. Taken together, these choices shape the economics of technology in an AI-first organization.

The imperative for CIOs is to act deliberately, not incrementally. They need to reassess hiring plans with a view of how roles will evolve, invest in internal capabilities to anchor long-term advantage, and reset vendor relationships according to outcomes rather than effort. For CIOs, none of these moves are easy and all require close collaboration with business leaders. But waiting for clarity carries its own risk. Organizations that fail to rebalance talent and sourcing now will find themselves locked into structures that limit both flexibility and returns tomorrow. CIOs who act decisively can shape the long-term economics of their technology organizations, rather than being limited by choices made too late.

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